

Chathumi Navodya Sathsarani

BSc in Applied Data Science Communication (Undergraduate)

+94 70 339 4005 wpcnsperera@gmail.com Hokandara East, Sri Lanka

linkedin.com/in/chathumi-navodya-sathsarani

github.com/ChathumiNavodya chathuminavodya.github.io/my-portfolio



Professional Summary

Final-year Applied Data Science undergraduate with hands-on experience in data analytics, machine learning, and data-driven solution development. Proficient in Python, SQL, and Power BI, with practical exposure to data cleaning, transformation, ETL processes, and interactive dashboard creation. Strong analytical and problem-solving skills with the ability to extract actionable insights from data. Seeking opportunities to grow in Data Analytics, Data Science, Data Engineering, and AI/ML roles.

Experience

Nawaloka Hospitals PLC

Junior Process Automation Analyst

Jun 2026 – Present

- Analyzed and documented hospital workflows across billing and departmental operations to identify process gaps, inefficiencies, and automation opportunities
- Designed process flow diagrams and business process documentation using Draw.io to support workflow standardization and digital transformation
- Supported the digitalization of hospital billing processes by analyzing manual activities, business rules, system requirements, and user workflows
- Developed Power BI dashboards and reports to support operational analysis and data-driven decision-making
- Used SQL and structured datasets to analyze operational information while maintaining data confidentiality
- Collaborated with business users and stakeholders to gather requirements, validate workflows, and identify opportunities for process improvement and automation

Sampath Bank PLC

Intern – Recoveries Department

Jun 2023 – Feb 2024

- Analyzed repayment patterns and overdue account data using Microsoft Excel to support decision-making processes
- Assisted in preparing reports and maintaining structured data for tracking recovery performance
- Supported daily operations by organizing and managing financial data efficiently
- Gained practical exposure to financial data analysis, risk monitoring, and workflow management

Education

Bachelor of Science in Applied Data Science Communication

2024 – 2027

General Sir John Kotelawala Defence University (KDU), Ratmalana

Certificate Course in Computer Applications

May 2023 – Nov 2023

University of Colombo, Sri Lanka

Certificate Course in English Language (Intermediate)

Apr 2023 – Oct 2023

University of Colombo, Sri Lanka

Skills

- **Programming:** Python, SQL, R
- **Machine Learning & AI:** Scikit-learn, TensorFlow, NLP, Model Evaluation, RAG Systems
- **Data Analysis:** Data Cleaning, EDA, Feature Engineering, Data Preprocessing
- **Data Engineering:** ETL Pipelines, Data Modeling (Star Schema)
- **Data Visualization:** Power BI, DAX, Plotly, Dash, Streamlit

- **Tools:** Git, REST APIs, SQLite, OCR, Microsoft Power Platform, Gemini API
- **Cloud:** Microsoft Azure
- **Microsoft Tools:** Excel, PowerPoint, Word
- **Soft Skills:** Communication, Teamwork, Problem Solving

Projects

- **AeroLogix AI – Gen AI Smart Airport Management System**  [GitHub](#)  [Live Demo](#)
Python, Streamlit, Machine Learning, Twilio API, LangChain, SQLite
Apr 2026
Developed an AI-powered airport dashboard to predict flight delays using machine learning, analyze real-time flight data, and send automated WhatsApp alerts to passengers.
- **Policy AI Assistant – Generative AI Policy Analysis System**  [GitHub](#)  [Live Demo](#)
Python, Streamlit, NLP, TF-IDF, Gemini API, HTML, CSS, JavaScript
Mar 2026
Developed a RAG-based AI system to automate policy document analysis with summarization and context-aware Q&A.
- **QuizCraft AI – AI Quiz Generation Platform**  [GitHub](#)  [Live Demo](#) Jan 2026
Python, Streamlit, SQLite, Gemini API, NLP, Data Visualization
Built an AI-powered quiz generation platform that converts documents into MCQs with analytics dashboards.
- **Flood Impact Prediction & Relief System**  [GitHub](#) Dec 2025
Python, Scikit-learn, Streamlit, Pandas, Matplotlib, Plotly
Developed an end-to-end machine learning system to predict flood occurrence and prioritize relief response.
- **UK Visa Sponsorship Analysis**  [GitHub](#) Nov 2025
SQL, Power BI, DAX, Data Modeling
Designed a star schema data model and developed interactive Power BI dashboards using advanced DAX measures.
- **Fake Job Detection System**  [GitHub](#) Oct 2025
Python, NLP, TF-IDF, OCR, SVM, SHAP, Streamlit
Built an AI-powered fraud detection system that analyzes job postings and provides explainable predictions.
- **Tourism Data Clustering System**  [GitHub](#) May 2025
R, K-means Clustering, ggplot2, dplyr, factoextra, Data Preprocessing
Performed clustering analysis on tourism data to segment user behavior and generate insights.
- **Student Performance Prediction Dashboard**  [GitHub](#) Apr 2025
Python, Pandas, Scikit-learn, Mlxtend, Plotly, Dash
Built a data pipeline combining association rule mining and logistic regression to analyze student performance.

Certifications

- **AI/ML Engineer – Stage 2** — SLIIT Jan 2026
- **AI/ML Engineer – Stage 1** — SLIIT Aug 2025
- **Data Analysis with Python** — IBM (Coursera) May 2025
- **GenAI 101 with Pieces** Feb 2025
- **Postman API Fundamentals Student Expert** Jan 2025

Leadership

AI and Data Science Club, KDU 2025 – 2026

Treasurer & Project Chairperson

- Managed financial records and budgeting for club events and initiatives
- Led planning and coordination of the **Tech Spark** event
- Collaborated with team members to organize and execute technical sessions

References

Ms. Chamodhi Wickramasinghe

Lecturer

General Sir John Kotelawala Defence University

☎ +94 71 361 2792

✉ wickramasinghe.bct@kdu.ac.lk

Dr. Charith Silva

Senior Lecturer

General Sir John Kotelawala Defence University

☎ +44 79 20 206 338

✉ charithsilva@kdu.ac.lk